CONCEPT GUIDE

Understanding Health Information Exchange (HIE)

What HIE is, why it matters, and the three ways patient data is exchanged

Overview

Health information exchange (HIE) is the secure, electronic sharing of patient health information among healthcare professionals, organizations, and, increasingly, patients themselves. HIE replaces the older model of paper records stored at individual medical offices, which made it difficult for providers to access a patient's full health history.

Because patient health information is confidential, every HIE transaction must follow your organization's privacy and security policies, including HIPAA requirements.

Why HIE matters

Giving providers fast, reliable access to patient data improves care in several ways:

- Reduces hospital readmissions.
- Helps prevent medication errors.
- Improves the speed and accuracy of diagnosis and treatment.
- Reduces duplicate testing.

Note: This guide assumes a working knowledge of basic healthcare IT terms. It's intended for new staff, students, or stakeholders who need a clear introduction to HIE before working with platforms that support it, such as Greenway Health.

The three types of HIE

HIE happens in three ways, depending on who needs the data and how urgently they need it.

Exchange type	How it works	Typical scenario
Directed exchange	Sends patient information securely from one known, trusted provider to another — similar to a secure email.	Sending lab results, referrals, or discharge summaries to a specialist.
Query-based exchange	Lets a provider search for and retrieve a patient's available clinical records on demand.	An emergency room physician retrieving a patient's history during an unplanned visit.
Consumer-mediated exchange	Gives patients direct access to and control over their own health information.	A patient reviewing their records online, similar to checking a bank account.

Directed exchange

Directed exchange lets a provider send patient information directly to another provider they already know and trust — for example, lab orders and results, referrals, or discharge summaries. The data travels over an encrypted, secure connection, similar to sending a secure email. Because the sender and recipient have an established relationship, directed exchange supports coordinated, ongoing care.

Example: A clinic sends quality-measure reports to the Centers for Medicare & Medicaid Services (CMS), or sends vaccination records to a public health agency.

Query-based exchange

Query-based exchange lets a provider search for and retrieve a patient's available clinical records when care is unplanned or urgent. This is the model most often used in emergency or first-encounter situations, where the treating provider doesn't yet have the patient's history on file.

Example: A physician treating a prenatal patient in the emergency department uses query-based exchange to retrieve her pregnancy care record, supporting better-informed decisions for both patient and unborn child.

Consumer-mediated exchange

Consumer-mediated exchange gives patients direct access to their own health information and control over how it's shared among their providers. It works much like online banking: patients can review their records and manage their care from anywhere.

Example: A patient managing a chronic condition logs in to a patient portal to review recent lab results and share them with a new specialist.

Summary

HIE moves patient information from paper-based, siloed records to secure digital exchange among providers and patients. Each of the three exchange types serves a different need: directed exchange for trusted provider-to-provider sharing, query-based exchange for unplanned or urgent care, and consumer-mediated exchange for patient-controlled access. Together, they support faster, safer, and more coordinated care while keeping patient data secure.